QA REQUIREMENTS FOR ADANI SOLAR PROJECTS

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

CONTENTS

SL. No	Description	Page No
01	OBJECTIVE	03
02	ABBREVIATION	03
03	RESPONSIBILITY	03
04	KICK-OFF / PRE-INSPECTION MEETING	04
05	PROCUREMENT OF MATERIALS AND INSPECTION METHODOLOGY	04
06	INSPECTION MANAGEMENT	05
06 (I)	SOURCE INSPECTION MANAGEMENT	05
06 (II)	DOCUMENTS REQUIREMENT FROM BIDDER	06
06 (III)	INSPECTION & MDCC ISSURANCE WITH FLOW CHART	07
06 (IV)	FIELD INSPECTION MANAGEMENT	08
06 (V)	QA MANPOWER & LAB SETUP	09
06 (VI)	INSPECTION & TESTING	10
06 (VII)	NDT REQUIREMENTS	10
06 (VIII)	POSITIVE MATERIAL IDENTIFICATION TESTING REQUIREMENTS	12
06 (IX)	GENERAL WEILDING REQUIREMENTS	13
06 (X)	GENERAL PRESSURE TEST REQUIREMENTS	14
06 (XI)	PAINTING REQUIREMENTS	15
06 (XII)	CALIBRATIONS	18
06 (XIII)	CIVIL & MISCELLANEOUS STRUCTURAL WORK:	19
06 (XII)	PACKING & DISPATCH, STORAGE AND PRESERVATION	26
06 (XIII)	AUDITS	27
07	ENCLOUSERS	27

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

1. OBJECTIVE

The Objective of this specifications is to provide framework for the Bidder, Bidder's suppliers for quality management of entire works.
This Specification covers the general requirements that shall be met after award of order / during contract execution the assigned project.

The Bidder is expected to get clarified all queries in respect of interpretation of any portion of this document before tendering in writing to the OWNER.

2. ABBREVIATION

QA	Quality Assurance	TPIA	Third party inspection agency
BOQ	Bill of quantity	MTC	Manufacture test certificate
SQP	Standard quality plan	MDCC	Material dispatch clearance certificate
SFQP	Standard field quality plan	SMAW	Shielded metal arc welding
MQP	Manufacture quality plan	GMAW	Gas Metal arc welding
ITP	Inspection Test plan	GTAW	Gas tungsten arc welding
FQA	Field Quality Assurance	SAW	Submerged Arc welding
NDT	Non-destructive testing	WPS	Welding procedure specification
ICP	Inspection categorization plan	PQR	Procedure qualification record
KOM	Kick of meeting	DFT	Dry film thickness
C&I	Instrumentation & control	HAZ	Heat affected zone
FAT	Factory acceptance testing	SAT	Site acceptance testing
CJP	Complete joint penetration	PJP	Partial Joint Penetration

3. RESPONSIBILITY

It is Bidders prime responsibility to arrange/produce the product conforming to contract specifications/ approved drawings/ Datasheet and inspect all equipment, materials and works at various stages of execution as per the approved Quality Plans. Bidder and other involved agencies to give adequate confidence that the activities are performed as per agreed Quality Plans and approved work procedures. Bidder shall deploy an exclusive team of Quality Control Engineers and Supervisors for ensuring the quality of works executed at site on day-to-day basis. Verification by Adani representative at any stage shall not relieve Bidder of his responsibility towards quality of the product. Bidder shall deploy quality engineer/ Third party inspection agency for Supplier Quality check as well as Quality check at site.

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

Responsibility matrix

Documents type	Approving authority Discipline wise. (Civil/Mech/E&I)	Remarks
Engineering documents TQ, Design data, GAD, TDS, CATALOUGE, PR, MR, Technical deviation approval	Adani Engineering	As per SOP
Quality documents QA requirements, SQP, ICP, QA procedure, FAT and Dispatch Clearance through mail, SIMS MDCC, NCR, Audit, Vendor assessment, make list approval, QA Deviation approval, Site QA support for supplied material	Adani QA - HO	
Quality documents Site IMIR, SFQP, QA procedure, RFI & routine/ final inspections, MC clearance, walk through & SAT, commissioning, testing & handing over	Adani QA - site	
Expediting, Project & EHS	As per SOP	As per SOP

4. KICK-OFF / PRE-INSPECTION MEETING

Successful Bidder shall share the Quality plan, Proposed vendor list, Inspection categorization plan during Kick of meetings with Adani team to clearly understand the requirements and discuss any concerns before the start of the manufacturing process.

5. PROCUREMENT OF MATERIALS AND INSPECTION METHODOLOGY

a) Adani Approved Vendor List:

The Bidder shall procure materials from Adani approved vendors only. In case Bidder proposes to procure materials from non-approved vendors, Bidder must submit deviation request during tender stage with credentials for proposed vendor, reason for deviation with back up documents to be submit for review and further action by Adani QA. Any proposal for deviation from Adani approved vendors will not be entertained post award of contracts.

Approved vendor list documents no. **Annexure II**

b) Non-Billable Items:

Bidder shall ensure all non-billable items not covered in BOQ but procured for completion of packages (i.e. Gaskets, Fasteners, cable ties, paints, welding consumables, additional quantities etc.) being inspected at source and & take the pre dispatch clearance from Adani QA.

Bidders shall not dispatch the material directly to site considering non billable items. Inspection of the same shall be carried out as per approved categorization plan.

Bidder shall procure non billable items from Adani approved vendor list only. Non billable items shall not be used at site without clearance of IMIR by Adani FQA.

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

c) Inspection Categorization Plan:

Inspection shall be carried out as per Adani approved Inspection categorization plan. Bidder shall provide all items details in categorization plan format, same shall be finalized during kick off meeting. Inspection methodology as below.

Category A: Quality Plan approval by Adani QA & Physical inspection/ witnessing of Test/ Checks of the item by Adani's Inspecting Engineer / (or Adani Nominated Inspection Agency) along with main Bidders' representative prior to dispatch.

Category B: Quality Plan Approval by Adani QA. Physical inspection by Main Bidders Third-Party Inspection Agency (TPIA) & Review and acceptance of Inspection Reports and Test Certificates (to be furnished by Bidder) by Adani, prior to dispatch of equipment.

Adani Nominated Inspection agency.

- TUV INDIA, TUV Rhineland, BVIL, CEIL, Tata project, IRS, SGS, DNV, LRQA.

Category C: The Main Bidder shall submit the Certificate of Compliance (COC), MTC & photos with identification marks prior to dispatch of the material to facilitate issue of MDCC.

Note: All documents shall be reviewed and endorsed by main Bidder QA before submission to Adani QA.

Inspection Categorization plan doc. no. – **Annexure I**

d) Quality Plans:

Bidder shall follow & endorse Adani Standard Quality Plans (SQP) and Standard Field Quality Plans SFQP wherever Adani SQP/SFQP for items/ activities are not available, Bidder shall submit MQP/ITP/FQP for Adani QA review & approval.
SQP (Standard Quality Plans), SFQP (Standard Field Quality Plans)
MQP Manufacture Quality plan), ITP (Inspection and test plan)
Adani standard SQP master list **– Annexure III**
Applicable SQP/SFQP- **Annexure V**

6. INSPECTION MANAGEMENT

I.) SOURCE INSPECTION MANAGEMENT:

Cat. A - Activities which are identified for Witness or Hold Point, specific inspection call shall be raised by the Bidder to Adani in the requisite format 4 days in advance with below documents.

- **P.O. Copy**
- **Specification**
- **Approved Data sheets & Drawings**
- **Approved SQP duly signed & stamped in all pages.**
- Valid type Test report approved by Engineering, if applicable.

Cat. B - Call shall be raised by the Bidder to Adani for information only with prior intimation in the requisite format 4 days in advance with below documents.

- P.O. Copy
- Specification
- Approved Data sheets & Drawings

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

- Approved SQP duly signed & stamped in all pages.
- Valid type Test report approved by Engineering, if applicable.

Cat. C – No inspection calls for witness as well as signed QAP require for Adani approval. Bidder shall submit documents as per approved Cat. plan.
Bidder to follow OWNER SAP SIMS system, SIMS ID & Password will be provided to successful Bidder after award of contracts to raise inspection call online on SIMS portal for A, B & C category items.

II.) DOCUMENTS REQUIREMENET FROM BIDDER.

Bidder shall take the dispatch clearance during post order to provide below set of documents before material dispatched to site.

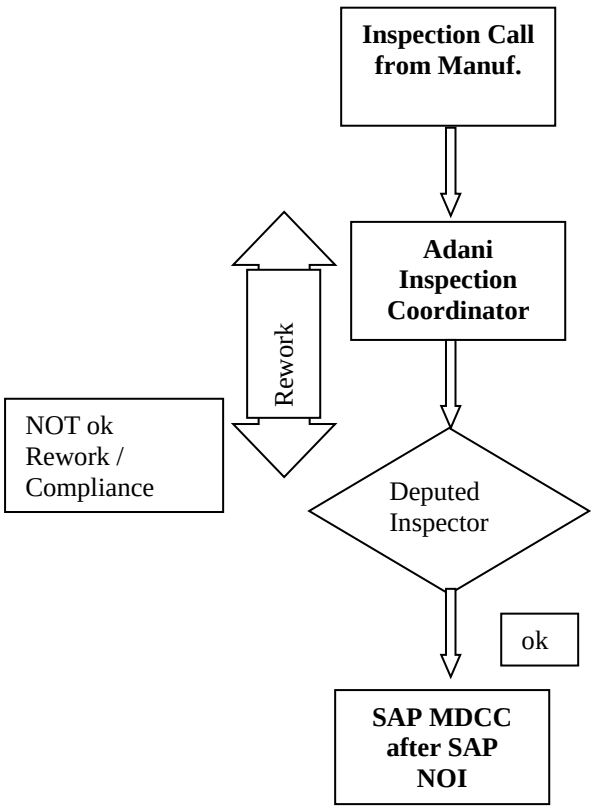
Supporting document submission	
Cat A inspected by Client/TPI	
1	Inspection waiver letter from client if the inspection is waived.
2	Inspection reports as per SQP stages
3	Approved SQP/MQP
4	Approved drawing/ Datasheet
5	Raw MTC, NDT reports
6	Painting requirements
7	Specific test report (i.e. hydro, functional test etc.)
8	SIMS call reference no
9	Photographs in QA dossier
Cat B Items QA dossier	
1	Main Supplier Inspection report
2	Approved manufacturer
3	QA dossier as per agreed MQP/SQP signed and stamped by Main supplier
4	Approved SQP/MQP of Supplier and reviewed(signed) by Client
5	Approved drawing/ Datasheet
6	Approved Painting requirement
7	Raw MTC, NDT report
8	Specific test report (i.e., hydro, functional test etc.)
9	Photographs in QA dossier
10	Raise SIMS call if all above applicable points are ok
Cat C Item QA dossier	
1	COC in client's format
2	TC/reports
3	approved drawing/datasheet
4	specific test report if applicable
5	Mark no in Identification column of COC
6	Raise SIMS call if all above applicable points are ok
7	Photographs in QA dossier
During Prebid stage	
1	Confirmation on QA requirements
2	Acceptance of ICP, SQP & SFQP

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

3	Acceptance of Vendor list, Painting specification
4	Owner acceptance of deviation (if any)
Agenda during KOM	
1	Sign off Categorization plan
2	Sign off SQP for CAT A & B
3	Sign off Vendor list if not in Prebid stage
4	Sign off SFQP
5	SIMS portal
6	Understanding the Inspection call process
7	Finalization on specific points if any
8	TPIA management

III.) INSPECTION & MDCC ISSUANCE WITH FLOW CHART

The inspection call shall be forwarded to Adani Quality inspection coordinator, communication matrix to be followed as per KOM.
The Quality representative will complete the inspection within the stipulated or scheduled date & time and the Inspection report shall be released with photographic evidence within one day of the date of inspection.



Sample Flow Diagram for ICMS

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

IV.) FIELD INSPECTION MANAGEMENT:

For the activities which are identified as Witness or Hold Point, RFI shall be raised by the Bidder to Adani in the requisite format 24 hours in advance.
Field Quality Plans (FQP) shall detail out for all the equipment, the quality practices, and procedures etc. to be followed by the Bidder "site Quality Control Organization", during various stages of site activities starting from receipt of materials/equipment at site till commissioning and final handover to OWNER.

In the Field Quality Plan, all stage inspection shall be classified into different class of check (i.e., A, B and C). "A & B" class of check shall be carried out in presence of OWNER's / OWNERS's representative and "C" class of check shall be witnessed by Bidder FQA Engineer at site. Work shall not proceed without witnessing of test by OWNER's/OWNER's representative for class "A & B" checks and Bidder representative for class "C" check.

The Bidder shall submit the Manufacturing and Field Welding Schedule (MWS & FWS) for field welding activities in the attached FORMAT for review and approval by OWNER. The field welding schedule shall be submitted to the OWNER along with all supporting documents like WPS, PQR, Heat treatment procedure, NDT procedures, etc. Bidder shall submit the schedule of submission of MWS & FWS along with the bid. Bidder shall do the quality planning for all the activities involved in deliverables. The quality planning shall cover as minimum the following:
Resources

Product/deliverables characteristics to be controlled. Process characteristics to ensure the identified product characteristics are realized.
Identification of any measurement & testing requirements and their acceptance criteria. Record to be generated and maintained.
Needs of any documented procedures.

The quality planning shall result into the quality assurance plan and job procedure for the project activities under Bidder scope. These documents shall be submitted to OWNER just after award of PO/contract and before start of work for review/ approval. OWNER may take minimum two weeks' time to review/approve these documents. Bidder shall validate any processes for production and service provision where resulting output cannot be verified by subsequent monitoring and measurement with no extra cost to the OWNER. This include any process where deficiencies become apparent only after the product is in use or service has been delivered.

Bidder shall deploy sub-Bidder as per PO/contract requirement only after going through stringent selection process and approval from OWNER.
Bidder shall establish a system for identification and traceability of product / deliverables throughout product realization. Product status with respect to inspection and testing requirements shall be identified and communicated to the OWNER from time to time. OWNER shall carry out a monthly Quality Audit of all equipment, Raw materials etc. lying in their store as per approved storage & preservation guidelines. The bidder shall establish and maintain a document to ensure that the product which does not confirm to the requirements is identified

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

and controlled to prevent its unintended use or delivery.

All non-conformities (NCs)/ deficiencies found by the Bidder inspection/ surveillance staff shall be dully recorded including their disposal action and shall be submitted to OWNER.
Product release or service delivery shall not proceed until the planned stage inspection is completed satisfactorily and recorded on prescribed format unless and otherwise approved by OWNER. No material shall be dispatched from the manufacturer's works before the same is accepted, after pre-dispatch and final inspection including verification of records of all previous tests/inspection by OWNER's representative or their authorized inspection agency and authorized for dispatch by issuance of Material Dispatch Clearance Certificate (MDCC).

Product requires re-assembly at project site shall have clearly identified matchmark at Bidder works before dispatch. Bidder shall ensure the details of match mark are recorded and sent along with packing list as well as in E-mail copy.

Bidder to submit Standard Operating Procedure (SOP) complying SFQP/ SQP/ FQP/ Tech Spec., for each activity, for OWNER's review and approval prior to start of work at site. OWNER's execution and QA team shall jointly review and approve the SOPs.

Bidder to comply total quality requirements as identified in SFQP/ SQP/ FQP/ Tech Spec. In the event of repetitive non-compliance, suitable penalty shall be imposed, and the amount shall be deducted from Running Account Bill (RA Bill).

V.) QA MANPOWER & LAB SETUP

The Bidder shall submit the details of the proposed FQA set up (organizational structure and manpower) with experience (minimum 6 to 7 years on similar kind of job) and skill for OWNER's approval. Bidder shall mobilize their FQA engineers at site well before start of erection activities.
Bidder shall establish and maintain full flagged Civil, Mechanical Lab, electrical and C&I with duly calibrated instruments from NABL accredited lab inside the boundary of project. Where no standards are available for calibration of measuring instruments, the basis used for calibration shall be recorded.
The deployment of experience / skilled QA manpower and site lab setup shall be linked with mobilization advance of successful Bidder and subsequent deduction shall be made from the running bills for every breakdown / non usable condition of equipment and non-availability of agreed QA manpower (quantity and quality) on daily basis.
Bidder shall appoint QA engineers at site for site quality check and ahead office for supply items inspections.
Mechanical, Electrical & Civil engineer: Must have minimum post qualification experience in the relevant field. Mechanical engineer shall have NDT level-II certificate in the relevant NDT method.

QA/QC engineer	Contract value	≤ 2 Crore	> 2 Crore
	Degree Holder	5 year	8 year
	Diploma Holder	8 year	10 year

Bidder shall submit their organization chart during pre-bid stages.

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

VI.)INSPECTION & TESTING

Bidder shall plan the inspection activity well in advance only after ensuring the compliances of all the requirements mentioned herein this document and inform to OWNER through request for inspection (RFI) / Inspection call format at least 48 hours before the inspection is to be performed at site and at least one week for inspection at vendor's work.
All equipment/ items being supplied shall conform to type tests including additional tests if any required as per technical specification and applicable National/International Standard. The type test certificate conducted earlier on similar type and rating of equipment not earlier than 5 (five) years from the date of award shall be submitted to Employer for review. In case the Type test is specially required as per Contract/PO/ Tech Spec, the same shall be performed by the Bidder and their sub-Vendors. In addition to the above all equipment / items being supplied shall be subject to routine & acceptance tests in accordance with requirements stipulated under respective sections in the specification, applicable codes and standard, manufacturer's plant standard and approved General Arrangement Drawing and Data sheet.

The particulars of the proposed FAT/SAT and the test procedures shall be submitted to the OWNER for approval prior to conducting the tests.
Welding consumable: Bidder shall use Adani approved brand & manufacture of welding consumable as per **Annexure IV**.

VII.) NDT Requirements:

The submission schedule for NDT procedures to be submitted by Bidder shall be finalized during Kick off meeting with the Supplier.
Bidder shall ensure NDT requirements at site during Etraction.

PROCESS PIPING as per ASME B31.3					
Service	**Material**	**Type of Examination**	**Type of Weld**		
			Butt Weld	**Socket Weld**	**Branch Weld**
CAT D Fluids (Cooling water, Raw water, CAS, CDA)	CS	a) Visual	100%	100%	100%
		b) LPT		10%	10%
		c) RT	10%		
	SS	a) Visual	100%	100%	100%
		b) LPT		20%	20%
		c) RT	20%		
	Non-Metallic	Visual	100%	100%	100%
Services - Other than CAT D Fluids	CS	a) Visual	100%	100%	100%
		b) LPT		100%	100%
		c) RT	100%		
	SS	a) Visual	100%	100%	100%
		b) LPT		100%	100%
		c) RT	100%		
	Non-Metallic	Visual	100%	100%	100%

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

Notes:

1) Branch welds shall consist of the welds between the pipe & reinforcing element (if any), nozzles & reinforcing element and the pipe & nozzle under the reinforcing element. Reinforcing element to be interpreted as pads, saddles, weldolets , sockolets etc.
2) Seal welds of threaded joints shall be given the same examination as socket welds.
3) Liquid Penetrant method of examination shall be in accordance with Section V of the ASME Boiler and Pressure Vessel Code, Article VII and V respectively. The entire area of the accessible finished weld surface shall be examined. Selected root runs, subject to a maximum of 10%, For normal services (Cat D) & 100% for other than CAT D services, before finished weld, may also be examined.

Radiography – For welds, each welder's and welding operator's work shall be represented.

Random Joint Percentage	Minimum Joints	Remarks
5%	Not less than one from each 20 welds	5% weld joints of each welder shall be represented.
10%	Not less than one from each 10 welds	10% weld joints of each welder shall be represented.
20%	Not less than one from each 5 welds	20% weld joints of each welder shall be represented.
More than 20%	Like Wise	Like Wise

Welds selected for examination shall not include flange welds and shall be radiographed for their entire length.

Structures			
NDT requirements	UT	RT	PT/MPT
Groove Weld (CJP)	--	100%	100% - root weld & final weld
Fillet weld (PJP)	--	--	100% - root weld & final weld
Lamination check -ASTM A435/578 level B	100%	--	PT at Edges

Equipment			
NDT requirements	UT	RT	PT/MPT
Groove Weld (CJP)	--	100%	100% - root weld & final weld
Fillet weld (PJP)	--	--	100% - root weld & final weld
Lamination check -ASTM A435/578 level B	100%	--	PT at Edges

Note:

1.) Unless otherwise specified in the approved drawing/ SQP
2.) Any Branch / Fillet weld after hydro test needs Adani quality approval. Any rework or rectification (Hot work) after hydro testing 100% Root DP & RT will be applicable.

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

VIII.) POSITIVE MATERIAL IDENTIFICATION (PMI) TEST REQUIREMENTS.

PMI to be carried out 100% at shop inspection & site during erection/ IMIR.

MOC	Frequency	Remarks
Stainless steel	100%	Weld, Base metal, HAZ
Alloy steel	100%	Weld, Base metal, HAZ

.

Sl.No	Item	Extent of PMI
1	Weld joints	100%
2	Rotary & static equipment	100%
3	Fasteners & supports	10%
4	Pipe, Fittings, Flanges, Valves	100%

a) The acceptable instruments for alloy analyzer shall be either "Portable X-Ray fluorescence" or "Optical Emission" type each capable of verifying the percentage of alloy elements within specified range.

b) Instruments or methods used for PMI examination shall be able to provide quantitative recordable, elemental composition results for positive identification of alloying elements present.

c) The method used for PMI examination shall provide a quantitative determination of the alloying elements like Chromium, Nickel, Molybdenum or Vanadium in Alloy Steel items.

d) Each analyzer must be calibrated according to the manufacturer's specification at the beginning and end of each shift. Instrument must be checked against known standard for each alloy type to be inspected during the shift.

e) The surfaces to be examined shall be prepared by light grinding or abrasive paper and solvent cleaner. Evidence of Arc burn resulting from examination shall be removed by light grinding or abrasive paper.

Acceptance Criteria:

1) **Base Metal:** PMI test results showing presence of characteristic elements up to 10% less than the minimum specified value in the material specification and up to 10% more than the maximum specified value in the material specification shall be acceptable.

2) **Deposited Weld Metal:** For deposited weld metal using welding consumables matching with base metals, the recorded presence of characteristic elements up to 12.5% less than the minimum specified value in the welding consumable specification and up to 12.5% more

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

than the maximum specified value in the welding consumable specification shall be acceptable.

Note-Read Adani Standard NDT Procedure (QA/Pro/04) for details.

IX.)General Welding requirements

Manufacturer shall need to get approval for WPS / PQR and shall get all its welder approved through WPQ/WQT/WOQ.

a) Reference Code- ASME Boiler and Pressure Vessel Codes
   Section II Part C - Specification for Welding Rods, Electrodes, and Filler Metals
   Section V - Non-destructive Examination
   Section IX - Qualification Standard for Welding and Brazing Procedures, Welders, Braziers, and Welding and Brazing Operators
b) ASME B 31.3 - Process Piping
c) Project Specification
d) Manufacturer Specification (For Non – Metallic Piping Only).
e) DVS 2207-11

Structural & Equipment's Welding Qualification Reference Code-ASME SEC-IX, AWS D1.1 Latest Edition

Type of Joints	MOC	Thickness	Process	Remarks
Grove Weld (CJP)	CS/ MS	-	SMAW / GMAW / FCAW /SAW	
Fillet Weld (CJP)	CS/ MS	-	SMAW / GMAW / FCAW /SAW	
STUD WELD	CS	All Dia	STUD WELDING	
Grove Weld (CJP)	SS	Less than 6.0mm thk.	GTAW	
Fillet Weld (PJP)	SS	More than 6 mm thk.	SMAW / GTAW	

Low hydrogen electrodes shall be properly baked.

Piping for Normal services requirements

Welding Qualification Reference Code-ASME SEC-IX, Latest Edition

Type of Joints	MOC	Root & Hot Pass	Fill-up & Final	Remarks
Grove Weld (CJP)	CS	GTAW	SMAW	
Fillet Weld (PJP)	CS	GTAW	SMAW	
Grove Weld (CJP)	SS	GTAW	GTAW	
Fillet Weld (PJP)	SS	GTAW	GTAW	

For Dia 24" & above Back chipping followed by seal run from inside.
SMAW shall be use for root welding provided the weld is back chip from the other side & sela run is applied.

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

Piping for Special services requirements

Welding Qualification Reference Code-ASME SEC-IX, Latest Edition

Type of Joints	MOC	Root & Hot Pass	Fill-up & Final	Remarks
Grove Weld (CJP)	SS	GTAW	SMAW	
Fillet Weld (PJP)	SS	GTAW	SMAW	

Tube Material for Special services requirements (Specialty Gases, High purity services).

Welding Qualification Reference Code-ASME SEC-IX, Latest Edition

Type of Joints	MOC	PROCESS	REMARKS
Grove Weld (CJP)	SS	ORBITAL WELDING	

Non-Metallic Pipes & Fittings requirements

Welding Qualification Reference Code-DVS 2207-11, Latest Edition

Type of Joints	MOC	Inter-pass & Final	Remarks
Grove Weld (CJP)	PP/PVDF/HDPE	BUTT FUSION	
Branch / Fillet Weld (PJP)	PP/PVDF/HDPE	SOCKET FUSION	
Double Containment outer Piping	HDPE	Electrofusion Coupling	Current & time as per Manufacturers Recommendation

Welding requirements for branch joint

1) The reinforcement pad shall be used from same pipe / same schedule and diameter of main pipe unless otherwise stated.
2) The reinforcement pad calculations shall be made in accordance with Engineering STD/ ASME B 31.3.
3) Branch joint 2" & below shall be welded by GTAW process only.
4) For Pipe-to-pipe branch connection shall be STUB IN type only.
5) For Equal size connection, Tee shall be used only (Stub in type connection)
6) For Connection type half coupling / weldolet / sockolet , stub on connection shall be applicable.

Note-Read Adani Standard Welding Procedure (QA/Pro/03) for details.

X.) General Pressure Test Requirements

Metallic Pipes, Tanks & Equipment's (Pressure Vessel, Heat Exchangers) (MOC-Carbon Steel, Alloy Steel, Stainless Steel) All pipes shall be required to comply the Pressure test requirements as per ANSI B 31.3

Testing	Requirement	Holding Time (Min.)	Acceptance Criteria	Remarks
Hydro Test	1.5 x Design Pressure	01 Hr	No Leakage & No Pressure drop	

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

Pneumatic Test	1.1 x Design Pressure	01 Hr	No Leakage & No Pressure drop	

Piping shall be properly cleaned & appropriate water quality (for stainless steel Chloride content for SS shall not be more than 50ppm) shall be used for hydro testing.

Non-Metallic Pipes, Ducts, Tanks & Equipment requirements (MOC-PP/PVDF/HDPE)

All pipes shall be required to comply the Pressure test requirements as per ANSI B 31.3

Testing	Requirement	Holding Time (Min.)	Acceptance Criteria	Remarks
Hydro Test	1.5 x Design Pressure	03 Hrs	No Leakage & No Pressure drop	
Smoke test/ Light test for ducts	No leakage	--	No leakage	Wherever hydro/Pneumatic is not possible with prior approval from Adani QA.

Piping shall be properly cleaned & appropriate water quality shall be used for hydro testing.

Torque Requirement for PVC Flange Joints:

Flange Size (in)	Torque (Kg-m)
1/2	1.383 - 2.073
3/4	1.383 - 2.073
1	1.383 - 2.073
1 1/4	1.383 - 2.073
1 1/2	1.383 - 2.073
2	2.764 - 4.416
3	2.764 - 4.416
4	2.764 - 4.416
6	4.563 - 6.915
8	4.563 - 6.915
10	4.563 - 6.915
12	4.563 - 6.915

Torque Requirement for HDPE/PP/PVDF Flange Joints:

Flange Size (in)	Torque (Kg-m)
1/2	-
3/4	-
1	-
1 1/4	-
1 1/2	-
2	-
3	6.500
4	4.149
6	6.224
8	7.192
10	7.745
12	8.851
14	9.128
16	10.373
18	16.596
20	19.362
22	22.128
24	24.894

Note-Read Adani Standard Pressure Test Procedure (QA/Pro/02) for details.

XI.)Painting Requirements

All External surfaces shall be C5M as per ISO 12944 Part5 / Adani Approved Colour code / Painting Procedure (unless otherwise specified in Approved Drawing. /Data sheet)
Paint Manufacturer
1.Asian Paints India Ltd

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

2. Shalimar Paints
3.Jotun
4. Akzonobel
5. Berger Paints
6. Good lass Nerolac Paints
7. Bombay Paints
8. Jenson & Nicholson

Pre-Qualification Test (PQT) shall be conducted and qualified by the manufacturer/supplier before first set of Primer painting which shall include (Pullout Test, Salt spray test, adhesion test etc.) from Approved Paint manufacturer with discussion with Adani quality team,
Table-A to be referred with Adani Painting Specification

TABLE-A

PAINTING REQUIREMENTS					
All External surfaces shall be C5M as per ISO 12944 Part5 / Adani Approved Colour code / Painting Procedure.					
Fabricated Structural Steel					
Scheme-01	Surface Preparation	SA2.5	No. Coat (Min.)	75-80 Surface profile	RAL
	Primer	Inorganic ethyl self-curing Zinc Silicate (SSPC-SP 20)	1	75 DFT	
	Intermediate	Super High build epoxy MIO coating cured with polyamide hardener	2	150 DFT	
	Finish Coat	High Build Gloss Aliphatic Acrylic Polyurethane	2	100 DFT	9002
		Total	5	325 DFT	
Scheme-02	Surface Preparation	SA2.5	--	75-80 Surface profile	
	Primer	Zinc Phosphate Epoxy	1	75 DFT	
	Intermediate	Zinc Phosphate Epoxy	2	150 DFT	
	Finish Coat	Polyamide cured colored pigmented	2	100 DFT	9002
		SUM	5	325 DFT	
Above Ground (AG) Piping) with Hot insulation					
Outside	Surface Preparation	SA2.5	--	75-80 Surface profile	
	Primer	Heat resistant silicone Aluminium Paint suitable up to 540°C dry temp.		--	

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

Above Ground (AG) Piping) with Cold insulation					
Outside	Surface Preparation	SA2.5	--	75-80 Surface profile	Adani Painting Spec. based on Service
	Insulation to be applied after primer application. Primer DFT 50-80 micron				
Above Ground (AG) Piping) without insulation (Except Fire & Sea Water Services)					
Outside	Surface Preparation	SA2.5	--	75-80 Surface profile	Adani Painting Spec. based on Service
	Painting shall be applied as per above Scheme-01 / 02				
Under Ground (UG) Piping for Normal Service (Except Fire & Sea Water Services)					
Outside Coating	Surface Preparation	SA2.5	--	75-80 Surface profile	
	Primer	Black / Zinc Phosphate	1	75-80 DFT	
	Coating	2 layers of 2.0mm Wrapping-Coating / Glass Flake epoxy 1.0mm thk	2	--	Holiday Test
AG Piping Sea & Fire Water Services,					
Inside	Surface Preparation	SA2.5	--	75-80 Surface profile	
	Coating	Coro Coat-Kirloskar / D-Mech (Chemglass 1000/Vinyl ester)	2	1000 Micron	Multiple Coats at field joint welds
Outside	Surface Preparation	SA2.5	--	75-80 Surface profile	
	Painting shall be applied as per above Scheme-01 / 02				Adani Painting Spec. based on Service
Total DFT				325 DFT	
UG Piping Sea & Fire Water Services,					
Inside	Surface Preparation	SA2.5	--	75-80 Surface profile	
	Coating	Coro Coat-Kirloskar / D-Mech (Chemglass 1000/Vinyl ester)	2	1000 Micron	Multiple Coats at field joint welds

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

Outside	Surface Preparation	SA2.5	--	75-80 Surface profile	
	COAT	2 layers of 2.0mm Wrapping-Coating / Glass Flake epoxy 1.0mm thk	2	1000 Micron	

All edges shall be protected from external damage & painting shall properly cured with desired DFT as per Approved Painting Procedure submitted by manufacture & approved by Adani Quality Team.
All components shall consist of Drawing No, Part/Item No, Rev., and the manufacturer duly stencilled for easy identification.
Any repair will take repair procedure approval from Adani Quality Team.

XII.) CALIBRATION

Bidder shall use below measuring instruments with NABL calibrated condition at site,

Civil		
Sr. No.	Descriptions	Frequency
1	Theodolite/ Total station	Once in a year
	Levels	Once in a year
	Measuring Tapes	Once in a year
	Cross staff	Once in a year
	Cube testing machine	Once in a year
	Moisture meter	Once in a year
	Batching plant	Once in 3 months
Mechanical		
	Measuring Tape/ scale	Once in a year
	Vernier calliper	Once in a year
	Ultrasonic thickness gauge	Before use
	Holiday/ spark tester	Once in a year
	DFT meter	Before use
	Weld gauge	Once in a year
	PMI machine	Before use
	Pressure gauge	Once in a year
	Portable Hardness testing	Once in a year
	Voltmeter	Once in a year
Electrical & Instrumentation		
	Ration meter	Once in a year
	Measuring instruments	Once in a year
	Temperature measuring instruments	Once in a year
	Vibration meter	Once in a year
	High voltage testing kit	Once in a year
	Insulation tester	Once in a year
	Digital Micrometer	Once in a year
	Digital Multimeter	Once in a year

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

	Sound level meter	Once in a year
Note:	All measuring instruments (other than above) must be calibrated once in a year.	

XIII.) CIVIL & MISCELLANEOUS STRUCTURAL WORK:

A. Earth Work:

i) BIDDER shall ensure all safety measures while excavation of soil / Blasting for hard rock and shall be carried out as per Approved Drawing requirement.

ii) Excavated soil shall be tested to identify the suitability of the soil as a backfill material. Testing shall be carried out in NABL accredited lab as per requirements stipulated in approved SFQP. If found suitable the same shall be stacked separately with proper identification.

iii) In case excavated soil is not suitable for backfilling OR not sufficient in quantity for backfilling, Bidder shall identify the Borrow area for soil in consultation with OWNER's Package Leads and FQA & shall be tested in NABL accredited lab as per requirement stipulated in Standard SFQP prior to start of Earth work. Tests shall be witnessed by OWNER's Execution / FQA.

iv) BIDDER shall use suitable mechanized equipment for compaction of approved soil in layers.

v) Proctor test (standard or modified) or relative density as per technical specification and drawings shall be carried out on all identified Soil as a backfill material.

vi) Backfilling must be carried out in required layer thickness as per specification / Standard SFQP. Log showing Reduced Level with Field Compaction Test result shall be maintained by BIDDER. If result of any compaction test failed to meet the requirement retesting of the same zone shall be carried out after proper compaction and should be reflected in the backfill logbook. BIDDER shall prepare layer thickness chart along with backfilling log and produce to OWNER during each filed compaction testing.

vii) Source of sand used for backfilling shall be approved from OWNER's Package Leads and FQA.

B. Concrete:

i) If the source of raw material (Cement, Coarse & Fine Aggregate, and Admixture) is identified by OWNER, BIDDER shall procure the raw material from the approved source only.

ii) In absence of approved source, the source of all raw material (Cement, Coarse & Fine Aggregate, Admixture) shall be identified by BIDDER based on sample collection from source, testing of the same as per approved SFQP in approved NABL accredited lab and compliance of the report as per acceptance criteria stipulated in Standard SFQP / Technical Specification / Relevant Codes & Standards. Sample collection & tests shall be witnessed by OWNER's Package Leads and FQA representative. Lab test reports shall be submitted to OWNER's Package Leads and FQA for approval.

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

iii) If Cement is Free Issue Material to BIDDER, source approval of the same is not required.

iv) Cement Storage facility at site must be created / established by BIDDER according to clause no. 4.2 of IS: 4082 (latest Revision).

v) Design Mix for concrete grades above & including M20 shall be performed at Third Party approved NABL accredited laboratory by BIDDER.

vi) BIDDER shall draw raw material samples in presence of OWNER's Package Lead and FQA from pre-approved sources and with the help of drawn samples minimum three numbers of Trial Mix of each grade of concrete, for each source of raw material shall be performed at approved NABL accredited lab. During this stage, BIDDER must ensure Admixture doses as per workability & retention time requirement at site, in consultation with OWNER's Package Leads and FQA. OWNER's FQA representative shall witness the casting of concrete cubes and testing of 3, 7 & 28 days of concrete compressive strength at approved NABL accredited lab.

vii) BIDDER shall submit final Concrete Mix Design of each grade of concrete along with proportion of all ingredients, & final product to OWNER's Package Leads and FQA for approval.

viii) Approved Concrete Mix Design shall be re-confirmed at site batching plant with minimum three trials for each grade of concrete, executed by BIDDER and witnessed by Owner FQA.

ix) In case of any change in source of raw material, BIDDER to carry out new Mix Design by taking Three Trials at Third Party NABL accredited Laboratory and at his site laboratory in presence of OWNER's FQA.

x) In case of change in brand of cement during the project cycle, BIDDER to perform Three Trials with new brand of cement and existing approved mix design of each grade of concrete at BIDDER's site laboratory to check the compatibility of cement and admixture in presence of OWNER's FQA. Based on the result of trial, BIDDER to adjust the doses of admixture in existing mix design. BIDDER to replace the existing admixture with new product or new brand of admixture in case of any compatibility issue arises between new cement brand and existing admixture during the trial.

xi) For mixing of concrete at site either weigh batch mixer machine with display screen or fully automatic batching plant shall be used by Bidder. In batching plant printing facility should be available to maintain records of each batch showing actual consumption of raw material by weight. Mixing of Concrete by volumetric batch is not permitted. Calibration of Batching plant shall be carried out by Bidder as per required frequency mentioned in Standard SFQP.

xii) Pre & Post Pour inspection shall be witnessed by OWNER as per Standard SFQP.

xiii) Filler material / Board & Sealant for Expansion / Construction joint in concrete shall be procured from OWNER's approved sources only.

xiv) Source & Type of Water Stopper shall be approved by OWNER. Water stopper shall be joint by adhesives or vulcanizing only.

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

xv) During Mass Concrete & Hot Weather concreting Log of Concrete Temperature shall be maintained by BIDDER at Batching Plant as well as at concrete pouring site.

xvi) NDT & Destructive Test like UPV Test, Rebound Hammer Test and Core Test shall be carried out by BIDDER whenever demand by OWNER's representative at no extra cost to OWNER.

xvii) Leak Test / Hydro Test for Water Retaining structures shall be carried out prior to Backfilling. Leak test shall be commenced after completing sufficient water soaking period of concrete i.e., min 72 Hrs. Rest of the procedure shall be followed as mentioned in this specification or as per Standard SFQP.

xviii) In case of any repair in concrete the same shall be carried out as per Pre-Approved Concrete Repair Procedure.

xix) Supporting formworks with the help of tie bar from rebar is not permitted. No binding wire and rebar shall protrude outside finished concrete unless desired by any specification or approved drawings.

xx) Shear keys shall be provided, and old concrete shall be hacked manually wherever necessary before placing new concrete. Construction joints shall be well prepared to SSD condition and Nitobond shall be applied before placing the new concrete on it.

C. Reinforcement:

i) Reinforcement steel Storage facility at site shall be created / established by Bidder according to clause no. 4.13 of IS: 4082 (latest Revision).

ii) Rusted reinforcement shall not be used until properly cleaned by wire brushing and dry hessian cloth.

iii) Bending of Reinforcement steel should be carried out by using correct mandrel size for respective dia. of steel. As per IS:1786, Mandrel dia. for FE 415 & FE 500 should be 4D & 5D respectively, where "D" is Diameter of Reinforcement Bar.

iv) If Reinforcement steel required to be welded, the same should be carried out as per IS: 9417 (Latest Revision).

v) Pre-Pour checks shall be carried out as per Class of Checks "A, B or C" mentioned in Standard SFQP to check dia., positioning, dimension, numbers of Reinforcement fixed in concrete element as per Bar Bending Schedule / approved drawing.

D. Formwork:

i) Source of Plywood / steel plate etc. going to be used as formwork shall be get approved by OWNER.

ii) Formwork of each structure shall be selected as per respective finishing requirement of the structure mentioned in this specification.

iii) To fix the formwork with each other Tie rods with threads at both ends covered with PVC pipe & cone shall be used by BIDDER.

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

iv) In water retaining structures Tie rod with Water Barrier shall be used to prevent any potential leakage from the concrete wall.

v) Pre-Pour checks shall be carried out as per Class of Checks "A, B or C" mentioned in Standard SFQP to check positioning, dimension, verticality of Formwork fixed for concrete element as per approved drawing.

E. Embedded Parts:

i) Source of all embedded parts shall be get approved from OWNER.

ii) Pre-Pour checks shall be carried out as per Class of Checks "A, B or C" mentioned in Standard SFQP to check positioning, dimension, level, verticality of Embedded Parts in concrete element as per approved drawing.

F. Masonry Work (with Bricks, Concrete Blocks, Rubble Masonry):

i) Source of Bricks, Concrete Blocks & Stone for Masonry work shall be got approved from OWNER as per the requirement of this specification and approved drawing.

ii) Soaking & continuous curing arrangement shall be done by BIDDER prior & after the masonry work.

iii) Source of Cement & Sand shall be approved from OWNER.

iv) Routine Testing of Bricks, Concrete Blocks, Stone and Cement Mortar shall be carried out as per frequency stipulated in Standard SFQP.

v) Storage of Bricks & Concrete Blocks shall be as per Clause 4.5 & 4.6 of IS: 4082

vi) Masonry work shall be get inspected after completion as per Class of Check "A, B or C" mentioned in Standard SFQP.

G. Plaster Work:

i) Plaster shall be carried out as per the required type for respective location stipulated in this specification or in Approved Drawing.

ii) Field Inspection of plaster shall be carried out at three stages i.e. Prior to start plaster, During Plaster & after plaster as per Class of Checks "A, B or C" mentioned in Standard SFQP.

iii) In case of any deficiency the same shall be rectified immediately by approved repair procedure.

H. Fabrication & Erection of Structure:

i) All MS angles, flats, box, etc. shall be procured from approved source only.

ii) WPS & PQR shall be prepared and got approved from OWNER prior to start fabrication at site.

iii) Welding consumables going to be used for fabrication shall be procured from list of "Recommended welding consumables for field welding" attached herewith.

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

iv) All prefabricated items (PEB) shall be reinspected for straightness, sweep, camber, dent, and damage before erection at site and joint protocol shall be maintained.

v) Foundation handing over shall be done from civil to mechanical well in advance.

vi) Fabrication and erection shall be carried out as per Standard FQP.

vii) Structure shall be aligned within ± 5mm. However, ± 10 mm maximum may be permitted in few cases.

viii) All butt joints in tension zone shall be 100% RT/PAUT tested.

I. Painting Work:

i) Brand / Type / Color shade for the painting shall be approved from OWNER.

ii) Inspection for surface preparation, putty application, first & second coat application of paint shall be carried out as per the class of checks mentioned in Standard SFQP.

iii) Color variation, patches shall be rectified prior to final handing over.

iv) Paint drums shall be stored under shed to avoid direct sunlight.

J. Grouting of Structural Columns & Equipment:

i) Type (Epoxy / Cementitious) & Source of grout shall be approved from OWNER.

ii) Grout material shall be approved as per the strength and other property requirement as mentioned in this specification and as per requirement of approved drawing.

iii) Proper Hacking & soaking of concrete surface should be completed prior to grout.

iv) Prior to start grouting activity "Grout Clearance" format shall be get signed from concerned authority (Mechanical OR Electrical Dept.) by BIDDER.

v) Not more than 4 packer plates shall be used under base plate of any structure column and equipment base frame.

vi) Pre & Post grouting inspection shall be carried out as per the class of check mentioned in Standard SFQP.

vii) Grout cube shall be cast & tested as per frequency stipulated in Standard SFQP.

K. Flooring Work:

i) Inspection & Test procedures (ITPs) for specialized flooring i.e., IPS with iroite toppings, IPS with Non-metallic hardeners toppings, Pre-cast terrazzo tile, Acid resistance tile, In-situ terrazzo, Ceramic tile, Carborundum tile, PVC tile or paver block tile as applicable, shall be submitted to Package lead and FQA for final approval.

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

ii) Source/ colour of stone (Marble, Kota stone etc.) & Brand/ Make/ Colour of Tiles shall be approved from OWNER as per requirement of this specification and relevant IS code.

iii) Storage of Tiles & Stones shall be properly arranged by BIDDER complying IS: 4082.

iv) Testing of Tiles & Stones shall be carried out as per frequency stipulated in Standard SFQP.

v) Slope, line, level & joint thickness shall be maintained as per approved drawing requirement and same shall be got inspected from OWNER as per class of check mentioned in Standard SFQP.

vi) Hollow sound test shall be carried out by OWNER as per frequency mentioned in Standard SFQP. In case of any deficiency observed the same tile / stone shall be re-fixed by bidder.

vii) Antiskid PVC tile shall be procured from pre-approved source only.

viii) Source of False flooring shall be approved from OWNER. Prior to commence False flooring activity Shop Drawing of the same shall be get approved from OWNER.

ix) Shop drawing shall include grid line, stud & base plate location of studs, location of full-size tiles & cut piece tiles according to size of room. It shall be prepared by bidder showing level of false floor.

L. Concrete Paving & Vacuum Dewater Flooring:

i) Prior to start Paving / VDF Field Density Test required to be carried out as per frequency mentioned in Standard FQP.

ii) Panel size / dimension shall be followed as per approved drawing and technical specification.

iii) For proper concrete finishing Power float for Paving / Power float & Vacuum pump for VDF shall be used.

iv) Filler material & Sealant for Construction / Expansion joint shall be get approved from Owner / Employer as mentioned in this specification / approved drawing.

v) Pre & Post pour inspection shall be carried out as per class of check stipulated in Standard FQP by OWNER.

M. Doors & Windows (Wooden, Steel, Aluminum, Fire doors):

i) Source of all types of Doors & Windows shall be approved from OWNER.

ii) For Fire rated doors Fire Rating test shall be carried out at NABL approved laboratory and report shall be submitted to OWNER.

iii) Storage of Doors & Windows shall be carried out as per IS 4820.

iv) Inspection like visual, dimensional, Paint thickness etc. of doors & windows shall be carried out prior to dispatch material from vendor premises or on receipt at site as per requirement of this specification / approved drawing / standard SFQP.

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

v) All accessories of Doors and windows shall be procured from approved source only.

N. Water Proofing:

i) Detail application procedure for water proofing shall be submitted by BIDDER for approval of OWNER.

ii) Source of material should be pre-approved prior to make any procurement.

iii) Stage wise inspection shall be carried out during application of Roof Water Proofing system by OWNER as per frequency stipulated in Standard FQP.

iv) Slope, line, level shall be properly maintained complying approved drawing requirement by bidder.

v) Leak test shall be carried out as per agreed procedure. Any dampness / leakage observed during leak test shall be rectified by bidder and shall be re-offered for leak test.

O. Plumbing works

i) Bought out material like Pipes, flanges, fasteners & gaskets shall be procured in accordance with the relevance standards. Test certificates shall be reviewed by OWNER.

ii) The complete piping loop shall be hydrostatically tested at 1.5 times the design pressure or 2(Two) times the working pressure, whichever is higher, for holding period of 30 minutes minimum and maximum up to the satisfaction of OWNER's inspection Engineer.

iii) All Pipes, fitting and other components shall be properly identified. Parts/ components requiring hydrostatic test shall not be painted/ concealed till hydrostatic test is performed satisfactorily.

Hydraulic test

A standard hydrostatic test shall be conducted at the pressure as per applicable standard on 100% of pipes by the manufacturer & hydro test of 10% of the pipes shall be witnessed by OWNER.

Repair

Any pipe which has been repaired after taking consent of OWNER shall be retested hydraulically at a pressure specified in the applicable standard.

P. Water Supply & Drainage system.

i) Source of all types of Pipes like GI, CI, PVC, UPVC, Stoneware etc. shall be approved from OWNER.

ii) Stage wise inspection as well as Hydro test, smoke test and Straightness test shall be carried out by OWNER as per frequency stipulated in Standard FQP complying relevant IS codes and this specification.

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

XIV.) PACKING & DISPATCH, STORAGE AND PRESERVATION

All items shall be protected from rust, corrosion, and mechanical damage during transportation, shipment, and storage.
Rust preventive used on Flanges, Fittings, Pipes & machined surface to be welded shall be easily removable with petroleum solvent and shall not be harmful to welding.
Ends shall be suitably protected, and the protectors shall be securely and tightly attached.

Material Type	Storage & Preservation requirements	Remarks
Structure material	Wooden sleepers (minimum 150 mm height)	
Metallic Pipes – CS, AS	End cap & Rust preventive coating on full body	
Non- Metallic Pipe	End Cover & tarpaulin cover full body	
Metallic pipe SS	End cap & tarpaulin cover full body	
Ducting	Proper support to be required during transit damage	
Flanges, CS, AS	Serration protection & Rust preventive full body	
Fittings- CS, AS	End cap & Rust preventive coating on full body	
Flange - SS	Serration protection & Wooden box	
Fittings - SS	End cap & Wooden box	
Valves	Flange end cover & Wooden box	
Static equipment, Pressure vessel, heat exchangers, Tanks, etc.	Engineering & OEM guideline	
Rotary equipment, Pumps, Compressor, fan, etc.	Engineering & OEM guideline	
Material Handling equipment	Engineering & OEM guideline	
Electrical and instrumentation equipment/ Items	Indoor storage on above ground/ OEM guideline	
Transformer, earthing material etc.	Outdoor storage on above ground/ OEM guideline	
Cements, grouting material, Paint, chemical	Indoor storage on above ground/ OEM guideline	
Imported items	OEM recommended & Sea-worthy packing	
Hazardous fluids, equipment & instruments	As per statutory guidelines	IBR, PESO, ATEX etc.

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

XV.) AUDIT

Bidder shall plan and carry out QMS audit for the job. Quality audit program shall cover design, procurement, construction, and commissioning as applicable including activities carried out by sub-vendors and sub-Bidders. This shall be additional to the Bidder ISO certification audit management program.
In addition to this OWNER shall perform quality and technical compliance audit. Any findings shall be immediately rectified by Bidder at no extra cost to OWNER.
OWNER reserves the right to involve in every audit as per his convenience with prior intimation to the Bidder. Bidder shall cooperate with full support and share all the reports for his record.
Quarterly Audits by Adani Quality Teams shall be planned each calendar year for verification of process & product.

7.) ENCLOUSERS

a.) General Inspection Categorization Plan - Annexure I
b.) Vendor list- Annexure II
c.) SQP SFQP index – Annexure III
d.) Welding consumable specification - Annexure IV
e.) SQP & SFQP - Annexure V
f.) Format - Annexure VI

We reserve all rights in this document and in the information therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.